

APPLICATION OF INFORMATION & COMMUNICATION TECHNOLOGIES

NETWORKING AND BLOCKCHAIN
INTEGRATION

RECAP OF LAST LECTURE

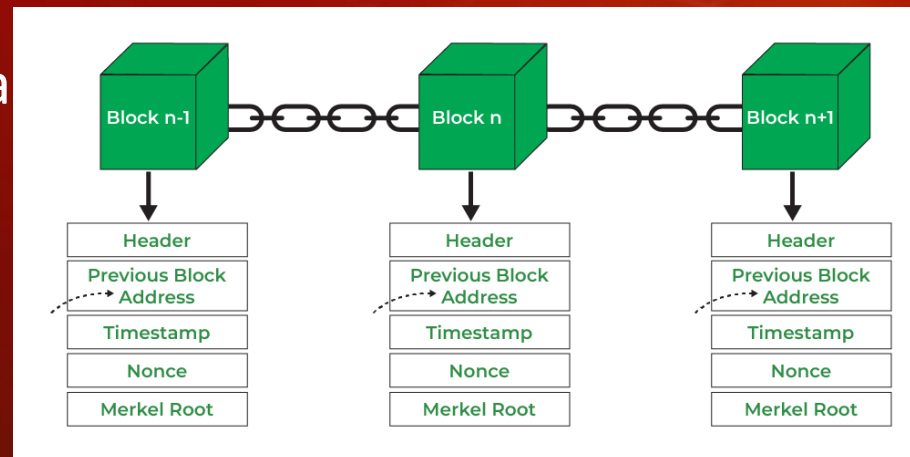
- Understand what Artificial Intelligence (AI) means
- Identify real-life uses of AI
- Differentiate between Narrow AI and General AI
- Discuss the benefits and challenges of AI
- Explain what Generative AI means
- Identify popular Generative AI tools
- Understand how to write good prompts
- Recognize ethical and safe use of AI

OVERVIEW OF CURRENT LECTURE

- Explain the concept of blockchain
- Describe how blockchain relies on computer networks
- Identify key applications and advantages of blockchain systems

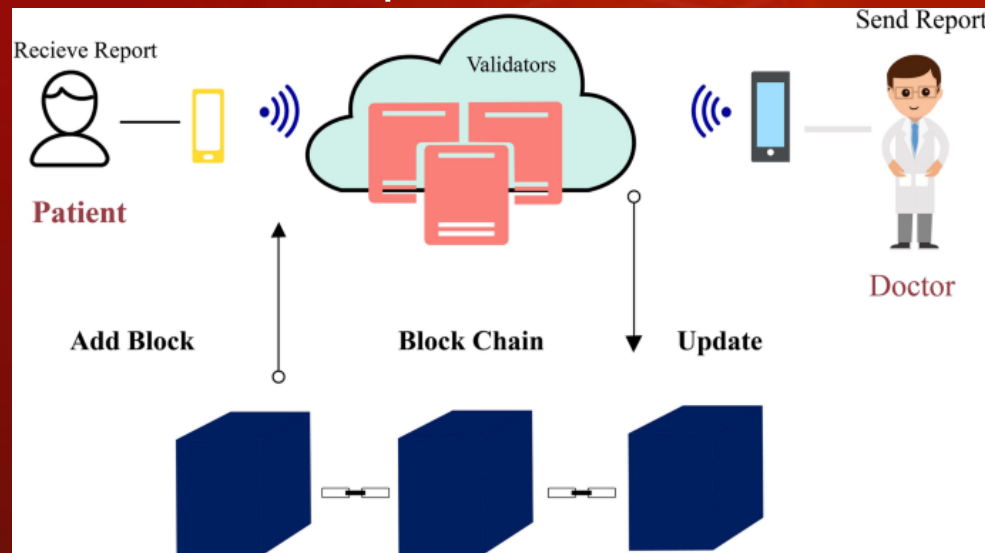
WHAT IS BLOCKCHAIN?

- A blockchain is a digital ledger that stores data in blocks, linked together in a chain.
- Each block contains:
 - Data (transactions, records, etc.)
 - A timestamp
 - A unique hash (digital fingerprint)
 - Hash of the previous block
- Once data is a



HOW BLOCKCHAIN WORKS

- A transaction is requested.
- The request is broadcasted to all computers (nodes) in the network.
- Nodes verify the transaction using algorithms.
- Verified transactions are added to a block.
- Each new block is linked to the previous one → forms a chain.



HOW BLOCKCHAIN USES NETWORKS

Aspect

Role of Network

Communication

Nodes share transaction data across the network.

Verification

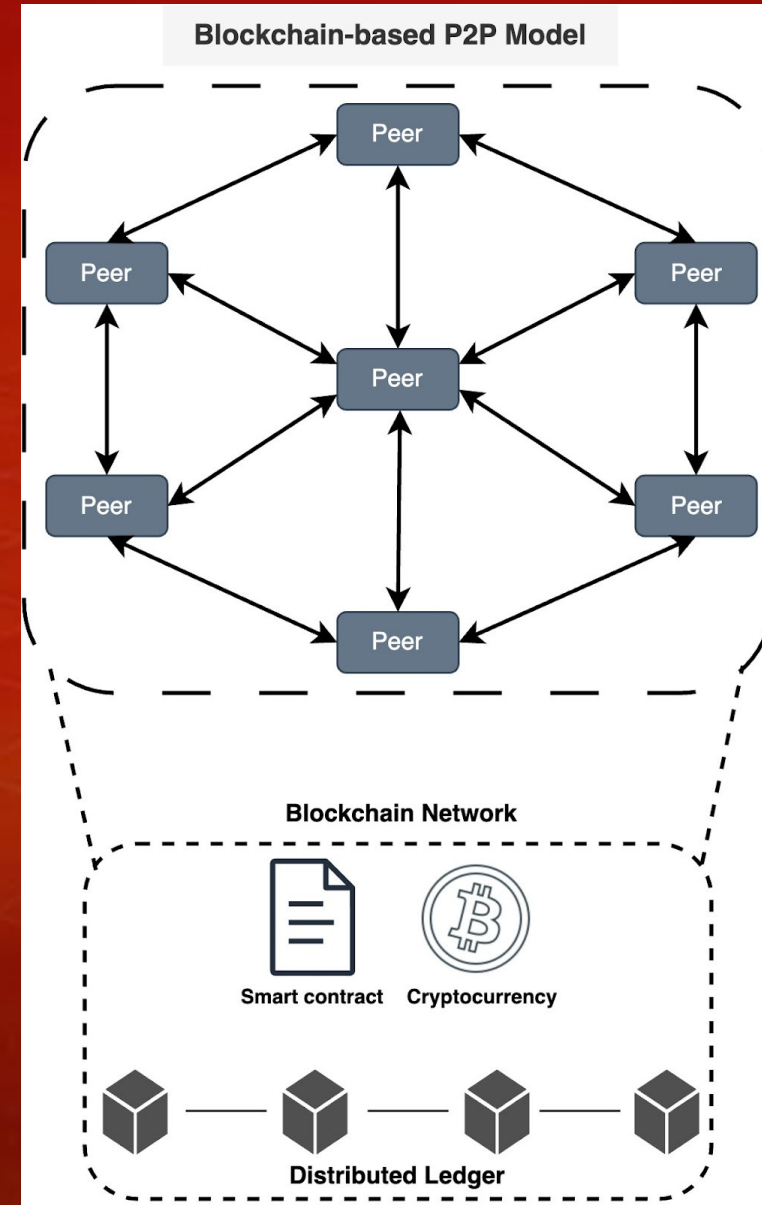
Distributed computers (peers) validate transactions together.

Storage

Data is stored across many devices (no single central server).

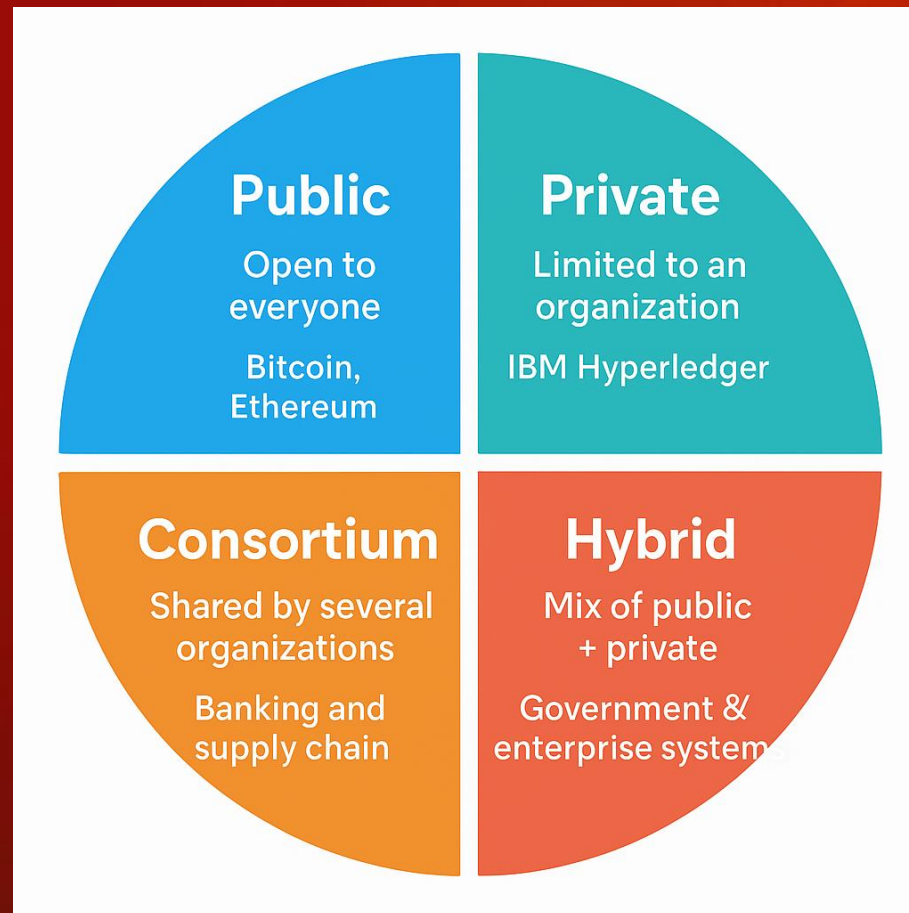
Security

Cryptographic linking and consensus make tampering almost impossible.



TYPES OF BLOCKCHAIN NETWORKS

Different blockchain networks provide varying levels of control, transparency, and decentralization.



APPLICATIONS OF BLOCKCHAIN

- Cryptocurrency: Bitcoin, Ethereum
- Smart Contracts: Automated digital agreements
- Supply Chain Tracking: Ensures authenticity of goods
- Healthcare: Secure patient data sharing
- Voting Systems: Transparent and tamper-proof
- Digital Identity: Protects user privacy



BENEFITS OF BLOCKCHAIN

- Transparency – everyone can verify transactions
- Security – cryptographic protection against tampering
- Decentralization – no single point of control
- Efficiency – faster settlement than traditional
- Trust – data integrity and traceability



- Blockchain ensures security and trust using encryption and distributed verification.

CHALLENGES / LIMITATIONS

- High energy usage (for mining)
- Scalability issues (slow processing for large systems)
- Legal & regulatory uncertainty
- Complex integration with traditional systems

SUMMARY

- Blockchain is a secure distributed ledger system.
- It relies on networks of nodes to verify and share data.
- It supports key applications like cryptocurrency, contracts, and data sharing.
- Future ICT systems integrate blockchain with AI and IoT for trust and automation.